O-RAN.WG7.OMAC-HAR.0-v05.00

Technical Specification

O-RAN White Box Hardware Working Group Outdoor Macrocell Hardware Architecture and Requirements (FR1) Specification

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Contents

List of figures

List of tables

Foreword

This Technical Specification (TS) has been produced by O-RAN Alliance.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1. Scope

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

version xx.yy.zz

where:

x the first digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).

y the second digit is incremented when editorial only changes have been incorporated in the document.

z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies system requirements and high-level architecture for the Outdoor Macrocell deployment scenario case as specified in the Deployment Scenarios and Base Station Classes document [1].

In the main body of this specification (in any "chapter") the information contained therein is informative, unless explicitly described as normative. Information contained in an "Annex" to this specification is always informative unless otherwise marked as normative.

2. References

2.1 Normative references

The following referenced documents are necessary for the application of the present document.

[1] ORAN-WG7.DSC.0-V04.00 Technical Specification, 'Deployment Scenarios and Base Station Classes for White Box Hardware'.

[2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[3] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception". (Rel-16.3.0)

[4] ORAN-WG4.CUS.0-R003-v12.00 Technical Specification, "O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification".

[5] Small Cell Forum, SCF222.10.02 5G FAPI: PHY API Specification, https://scf.io/en/documents/222_5G_FAPI_PHY_API_Specification.php (accessed on March 6, 2024)

[6] Small Cell Forum, SCF223.10.01 5G FAPI: RF and Digital Frontend Control API, https://scf.io/en/documents/223_5G_FAPI_RF_and_Digital_Frontend_Control_API.php (accessed on March 6, 2024)

[7] 3GPP TS 38.113: "NR: Base Station (BS) Electromagnetic Compatibility (EMC)

[8] O-RAN.WG1.OAD-R003-v09.00 , O-RAN WG1: "O-RAN Architecture Description" - v09.00

[9] O-RAN.WG6.CADS-v04.00, O-RAN WG6: "Cloud Architecture and Deployment Scenarios for O-RAN Virtualized RAN" – v04.00.

[10] O-RAN.WG4.MP.0-R003-v12.00 Technical Specification, "O-RAN Management Plane Specification" – v12.00.

[11] O-RAN.TIFG.E2ETSTFWK.0-v01.00 , "O-RAN End-to-End System Testing Framework Specification 1.0 " – v01.00.

[12] O-RAN White Box Hardware Working Group Outdoor Macrocell Hardware Reference Design Specification

[13] 3GPP TS 36.104: "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) radio transmission and reception". (Rel-17.6.0)

[14] O-RAN.WG11.Security-Requirements-Specification.O-R003-v06.00, O-RAN WG11, "O-RAN Security Requirements Specification 6.0 " – v06.00.

[15] O-RAN.WG11.Security-Protocols-Specification.O-R003-v06.00, O-RAN WG11, "O-RAN Security Protocols Specifications 6.0", - v06.00.

[16] O-RAN.WG11.Security-Test-Specifications.O-R003-v04.00, O-RAN WG11, "O-RAN Security Test Specifications 4.0", - v04.00.

[17] O-RAN.SuFG.CE.-v01.00, O-RAN SuFG, "Circular economy guidelines on network equipment – Technical Report", - v01.00.

2.2 Informative references

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

This section intentionally left blank.

This section intentionally left blank.3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in [2] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in [2]. For the base station classes of Pico, Micro and Macro, the definitions are given in [3].

Carrier Frequency: Center frequency of the cell.

F1 interface: The standard interface between an O-CU and an O-DUx specified in [8].

Integrated architecture: An architecture wherein the O-RU$_x$ and O-DU$_x$ are implemented on one platform. Each O-RU$_x$ and RF frontend are associated with one O-DU$_x$ and are connected to O-CU via F1 interface

Split architecture: An architecture wherein the O-RU$_x$ and O-DU$_x$ are physically separated from one another. A switch may aggregate multiple O-RU$_x$ (s) to one O-DU$_x$. The O-DU$_x$, switch and O-RU$_x$ (s) are connected by the fronthaul interface as defined in [4].

Transmission Reception Point (TRxP): Antenna array with one or more antenna elements available to the network located at a specific geographical location for a specific area.

Occupied Bandwidth (OBW): It refers to the bandwidth occupied by the base station when operated, defined by the sum of the active bandwidths of the band allocation(s) operated. As defined by 3GPP TS 34.121 section 5.8, occupied bandwidth is the bandwidth containing 99% of the total integrated power of the transmitted spectrum, centered on the assigned channel frequency. The bandwidth between the 0.5% power frequency points is the occupied bandwidth.

Instantaneous Bandwidth (IBW): It refers to the bandwidth in which all frequency components can be simultaneously analyzed. It is defined by the frequency boundaries of the operating band(s).

Frequency Range: It refers to bandwidth defined by the frequency range within which the Base Station can be operated, defined by the band-pass filter of the BS; e.g., 3.4 – 3.8 GHz (400 MHz)

gNB: A RAN node providing NR user plane and control plane protocol terminations towards the UE, and connected via the NG interface to the 5GC

Frequency Band: A designated frequency range for the operation of the base station and the UE radios. 5G NR frequency bands are divided into two different frequency ranges: Frequency Range 1 (FR1) that mainly includes sub-6GHz frequency bands, some of which are bands traditionally used by previous standards but has been extended to cover potential new spectrum offerings from 410MHz to 7125MHz; Frequency Range 2 (FR2) that includes frequency bands from 24.25 GHz to 52.6 GHz. Bands in this millimeter wave range have shorter range but higher available bandwidth than bands in the FR1.

Fronthaul Gateway (FHGW): A fronthaul gateway is a physical entity that is located between a distributed unit and one or more radio units where it distributes, aggregates, and/or converts fronthaul protocols between the distributed unit and multiple radio units.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

N_{RB} Transmission bandwidth configuration, expressed in units of resource blocks (for E-UTRA)

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

7-2x	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2
3GPP	Third Generation Partnership Project
5G	Fifth-Generation [Mobile Communications]
5GC	5G Core
ACS	Adjacent Channel Selectivity
ADC	Analog to Digital Converter
AFE	Analog Front End
AFU	Antenna Filter Unit
BF	Beamforming
BMCA	Best Master Clock Algorithm
BPSK	Binary Phase Shift Keying
BS	Base Station
C2C	Chip to Chip Interface
CAL	Calibration
CFR	Crest Factor Reduction
CLK	Clock
CPRI	Common Public Radio Interface
CU	Central Unit
DAC	Digital to Analog Converter
DDC	Digital Down Conversion
DDR	Double Data Rate
DFE	Digital Front End
DL	Downlink
DPD	Digital Pre-Distortion
DSP	Digital Signal Processor

DU	Distributed Unit
DUC	Digital Up Conversion
eCPRI	Enhanced Common Public Radio Interface
EMC	ElectroMagnetic Compatibility
EVM	Error Vector Magnitude
FAPI	Functional Application Platform Interface
FEC	Forward Error Correction
FFT	Fast Fourier Transform
FH	Fronthaul
FHGW	Fronthaul Gateway
$FHGW_{x \rightarrow y}$	Fronthaul Gateway that can translate FH protocol from an $O\text{-}DU_x$ with split option x to an $O\text{-}RU_y$ with split option y, with currently available option 7-2x→8.
FHM	Fronthaul gateway with no FH protocol translation, supporting an O-DU with split option x and an O-RU with split option x, with currently available options 6→6, 7-2x→7-2x, 8→8
FPGA	Field Programmable Gate Array
GbE	Gigabit Ethernet
GPP	General Purpose Processor
IEEE	Institute of Electrical and Electronics Engineers
IMD	InterModulation Distortion
IP	Internet Protocol
I/O	Input/Output
JTAG	Joint Test Action Group
L1	Layer 1, also referred to as PHY or Physical Layer
L2	Layer 2, also referred to as Data Link layer in OSI model
L3	Layer 3, also referred to as Network layer in OSI model
LLS	Lower Layer Split
LNA	Low Noise Amplifier
LTE	Long Term Evolution
MAC	Media Access Control
MIMO	Multiple Input Multiple Output
mMIMO	Massive Multiple Input Multiple Output
MCP	Multi-Chip Package
nFAPI	Network Functional Application Platform Interface
NG	Next Generation
NR	New Radio
OAM	Operations, Administrations and Maintenance
O-CU	O-RAN Centralized Unit as defined by O-RAN
O-DU	O-RAN Distributed Unit as defined by O-RAN
$O\text{-}DU_x$	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2x (as defined by WG4) or 8
O-RU	O-RAN Radio Unit as defined by O-RAN
$O\text{-}RU_x$	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2x (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the $O\text{-}DU_x$
PA	Power Amplifier
PCIe	Peripheral Component Interface Express
PDCP	Packet Data Convergence Protocol
PHY	Physical Layer, also referred as L1
PLL	Phase Locked Loop
POE	Power over Ethernet
QAM	Quadrature Amplitude Modulation

QPSK	Quadrature Phase Shift Keying
RAN	Radio Access Network
RF	Radio Frequency
RFIC	Radio Frequency Integrated Circuit
RFSoC	Radio Frequency System on Chip
RLC	Radio Link Controller
RRC	Radio Resource Controller
RU	Radio Unit as defined by 3GPP
RX	Receiver
SDU	Service Data Unit
SFP	Small Form-factor Pluggable
SFP+	Small Form-factor Pluggable Transceiver
SoC	System on Chip
SPI	Serial Peripheral Interface
TR	Technical Report
TS	Technical Specification
TX	Transmitter
UL	Uplink
USB	Universal Serial Bus
WG	Working Group

4 Deployment Scenarios and White Box Base Station Architecture

This chapter consists of two parts: the deployment scenario and the white box architecture. The deployment scenarios outline more specific functional requirements of the base station. All the reference designs shall meet these requirements in order to comply with O-RAN white box standard. In the white box hardware architecture section, it describes the overall gNB hardware architecture and function partition that meet the design requirements. The details on each of these topics are described in the following sections.

4.1 Deployment Scenarios

The outdoor macro-cell is the targeted deployment scenario of this specification. The requirements of the outdoor macro-cell base station are listed in the white-box Deployment Scenarios and Base Station Classes document [1]. Some of the key requirements described in that document are highlighted here.

Table 4.1-1 Deployment scenarios

Cell type	Outdoor Macro-cell
Frequency Band	FR1 (n26, n29, n41, n66, n70, n71, n77, n78, B1/n1, B3/n3, B7/n7, n78 CEPT(*)).
Instantaneous Bandwidth	Up to 205 MHz for FDD Up to 400 MHz for TDD
Occupied Bandwidth	Up to 205 MHz for FDD up to 300 MHz for TDD
Inter site distance	0.5 ~ 2 mile(s)
Antenna Configuration	4T4R ~ 64T64R
Fronthaul Type	O-RAN FH, Split Option 7-2x CAT-A and CAT-B Radio Units
Midhaul	Split Option 2

(*) n78 CEPT makes reference to an European n78 sub-band (according with ECC Decision (11) 06 and CEPT Report (67)

4.2 White Box Base Station Architecture

In general, the base station hardware architecture can be classified when O-$DU_{7\text{-}2x}$ is at cell site or O-$DU_{7\text{-}2x}$ is at the data center. In both cases, O-RAN fronthaul split 7-2x and 3GPP standard interface F1 will be used.

When O-$DU_{7\text{-}2x}$ is at the cell site, the O-RUs will be connected from a multi-sector cell site to the O-$DU_{7\text{-}2x}$. An optional switch/cell site router may be used between cell site O-RUs and O-$DU_{7\text{-}2x}$ (i.e. Scenario E.1 from WG6).

When O-$DU_{7\text{-}2x}$ is at a data center (i.e., the Edge Cloud from [9]), cell site routers/switches will aggregate O-RUs from multiple cell sites to the O-$DU_{7\text{-}2x}$. The O-$DU_{7\text{-}2x}$ will be connected to the O-CU-CP and O-CU-UP in the mid-haul through F1-C and F1-U interface. The O-CU-CP/O-CU-UP can be in a remote data center (i.e. the Regional Cloud in Scenario C from [9]) or, respectively the O-CU-CP/O-CU-UP can be co-located with O-$DU_{7\text{-}2x}$ in the data center (i.e. the Edge Cloud in Scenario B from [9]). Accordingly, when the O-$DU_{7\text{-}2x}$ is deployed at a data center site (i.e., on the Edge Cloud, either on a bare-metal machine or as a container/virtual-machine), the servers running the O-$DU_{7\text{-}2x}$ shall follow the existing and upcoming specifications from O-RAN WG6. The cell site cloud specifications are for further study in WG6.

The architecture when O-$DU_{7\text{-}2x}$ is at the cell site is shown in Figure 4.2-1. The O-$DU_{7\text{-}2x}$ with integrated switch/cell site router is considered for this architecture.

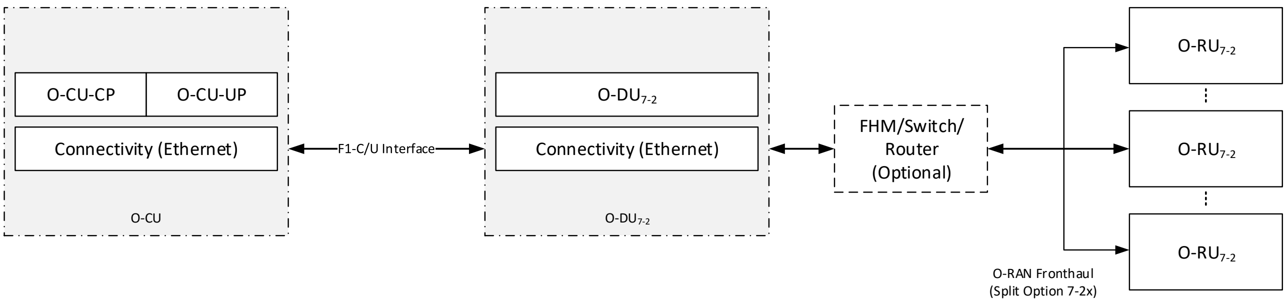

Figure 4.2-1: Outdoor macro-cell architecture with O-$DU_{7\text{-}2x}$ at cell site and data center

5 White Box Hardware Architecture

Based on the gNB physical implementation architectures discussed earlier, this chapter provides the architecture, major building blocks and all external/internal interfaces for each whitebox.

5.1 O-CU Hardware Architecture

In 3GPP system architecture, the gNB Central Unit (CU) communicates with the Distributed Unit (DU) via an F1 interface [8]. This interface has been adopted by O-RAN Alliance for O-CU and O-DU connection. F1 is an IP-based protocol interface. F1 interface is split into F1-C and F1-U interfaces.

F1-C interface is used for the management of interface setup/configuration, system information, UE context, RRC transfer, paging and warning message transfer. F1-U interface is used for user data transfer and flow control.

5.1.1 O-CU Architecture Diagram

The O-CU can be implemented with any General-Purpose Processor (GPP) based platform having an optional hardware accelerator block. The O-CU functions can be implemented either on separated hardware platforms or on the same hardware platforms integrated with O-DU$_x$ functions. In either case, the O-CU hosts L2/L3 functions, whereas O-DU hosts L2/L1 functions which require different CPU, Storage, Acceleration and Networking capabilities. Figure 5.1-1 shows the hardware blocks and interfaces within the O-CU white box.

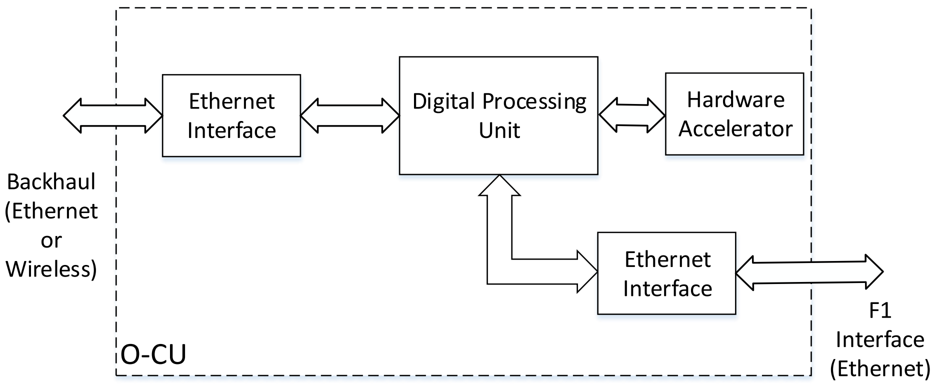

Figure 5.1-1: O-CU Hardware Block Diagram

5.1.2 O-CU Functional Module Description

The O-CU functional architecture comprises Digital Signal Processing, optional HW Acceleration, and Connectivity (GbE) units as well as memory and storage units as shown in Figure 5.1-1

5.1.3 O-CU Interfaces

The O-CU interfaces through backhaul with the vEPC or 5GC core network and via F1 interface with the O-DU$_{7\text{-}2x}$. The backhaul and the F1 interfaces are typically implemented with GbE transport/connections.

5.2 O-DU$_{7\text{-}2x}$ Hardware Architecture when deployed at cell site

Depending on the 3GPP standards and category of the radio unit, the split function blocks within O-DU$_{7\text{-}2x}$ and O-RU$_{7\text{-}2x}$ may vary accordingly. The O-RAN fronthaul C/U/S-plane specification [4] provides comprehensive information on this topic. The hardware functional partition architecture is shown in Figure 5.2-1. The High-PHY functions in O-DU$_{7-2x}$ are run on Digital Processing Unit, Hardware Accelerator, or both.

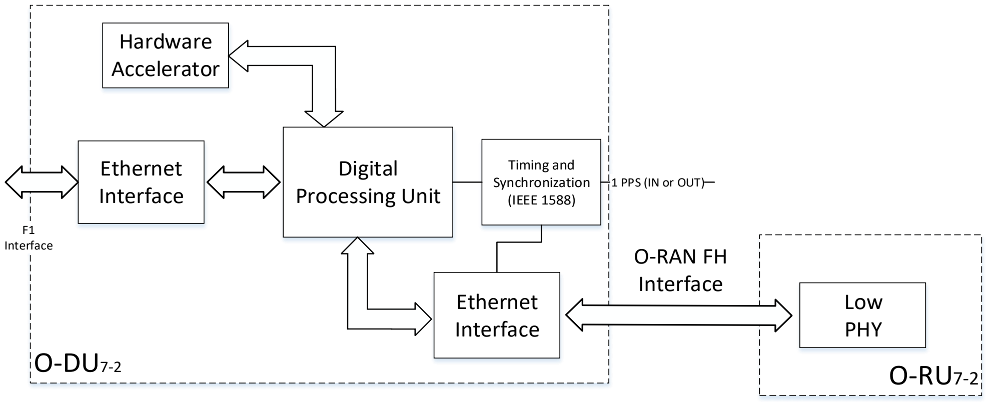

Figure 5.2-1: O-DU$_{7-2x}$ with split physical layer function

5.2.1 O-DU$_{7-2x}$ Functional Module Description

The choice of O-DU$_{7-2x}$ hardware components depends on the product design requirements and is outside the scope of this specification. Their descriptions are as follows:

Digital Processing Unit

The processing unit can be any GPP, FPGA or digital signal processor (DSP), with I/O chipset. It may also be in the form of System-On-Chip (SOC), or Multi-Chip Package (MCP).

Memory

DDR memory devices are used to store the runtime data and software for the processing unit.

Flash Memory

On board non-volatile storage device is used to store the firmware and non-volatile data, such as log data.

Board Management Controller

The controller is used to manage/control the power and monitors the operational status of the board.

Storage Device

The storage device such as hard drive is used to store OS, driver and applications software.

Ethernet Controller

The Ethernet ports transport the fronthaul or backhaul traffic according to the gNB hardware node requirements. The Ethernet device shall support IEEE1588 based timing synchronization.

Hardware Accelerator

The hardware accelerator is an optional device. For performance improvement, hardware accelerator can be used to process computationally intensive functions (e.g., Forward Error Correction (FEC)) and to offload the processor.

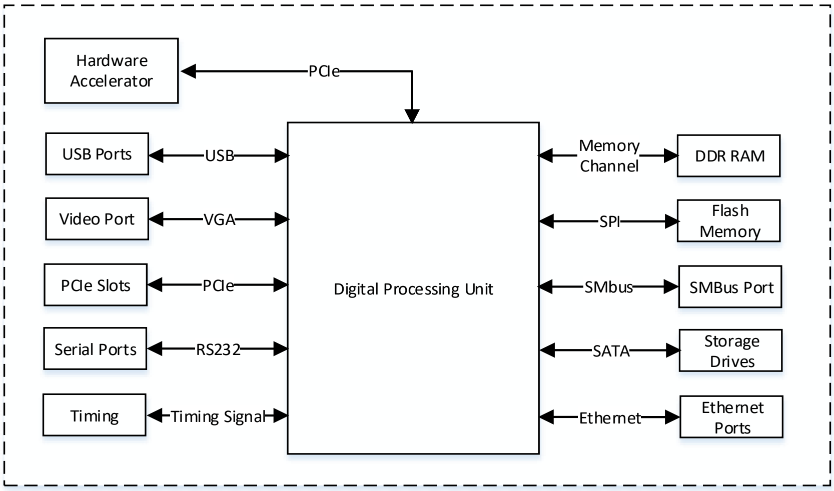

Figure 5.2-2: O-DU$_{7-2x}$ Functional Block and Interface Diagram.

5.2.2 O-DU$_{7-2x}$ Interfaces

The O-DU$_{7-2x}$ supported interfaces described below are also shown in Figure 5.2-2

Memory Channel Interface

Support DDR4 and later memory interface.

PCIe Interface

Support for PCIe Gen3 x16 and later interface versions; the bandwidth depends on the use cases, and it can be used to connect an accelerator device or network card.

Ethernet Interfaces

Supports any one or combination of 1GbE/10GbE/25GbE/40GbE links.

Serial ATA Interface

SATA3 shall be supported in case of software storage, such as hard drive.

SPI Interface

The SPI interface connects the processor with flash type of device for firmware.

Video Interface

Video interface is optional

USB Interface

Used to connect with local device for debug or on-board firmware update.

Miscellaneous Interface

Other interfaces that may be needed such as serial port, JTAG, etc.

5.3 O-DU6 Hardware Architecture when deployed at cell site

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5.4 O-DU8 Hardware Architecture when deployed at cell site

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5.5 FHM/Switch - Option 7-2x to Option 7-2x Hardware Architecture

The FHGW in the case of split option 7-2x to 7-2x is a multiplexer or a switch where the received data packets are distributed among or aggregated from one or more O-RUs and the signaling and management are appropriately routed to or from specific O-RUs.

5.6 FHM/Switch - Option 6 to Option 6 Hardware Architecture

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5.7 FHGW7-2x->8 - Option 7-2x to Option 8 Hardware Architecture

'This section intentionally left blank'. Additional details will be provided in a separate document from the fronthaul gateway task group.

5.8 FHM/Switch - Option 8 to Option 8 Hardware Architecture

'This section intentionally left blank'. Additional details will be provided in a separate document from the fronthaul gateway task group.

5.9 O-RU7-2x Hardware Architecture

In this section, high level architectures are presented for various O-RU configuration with split 7-2x. The configurations are nTnR O-RU where 'n' is commonly considered to be 4 or 8 for O-RU and 32 or 64 for mMIMO O-RU. In Section 5.9.1, a high-level functional block diagram is described for nTnR O-RU.

5.9.1 O-RU7-2x Functional Module Description

In Figure 5.9-1, we illustrate O-RU7-2x functional blocks that support O-RAN FH with split option 7-2x. There is at least one interface port which supports all fronthaul interface (for E.g., Optical Interface). For bandwidth reduction, O-RU7-2x architecture also supports the optional compression and decompression functions of FH interface. ASIC/FPGA based O-RAN Fronthaul Processing Unit of O-RU7-2x will receive the eCPRI packets and perform optical to electrical conversion, decompression, and security aspects according to WG11 specifications. Subsequently, the packets are sent to DFE through packet interface. The DFE block of O-RU7-2x is responsible for Low-Phy functions such as FFT/IFFT, CP addition/removal, and PRACH filtering. Additionally, DDC, DUC, CFR and DPD functions are also performed in DFE unit. The ADC and DAC functions are performed in the AFE, which are mixed signal blocks responsible for conversion of data between the digital to analog domains and vice versa. These signals from the AFE are fed to the Tx/Rx chains. The RF Processing /RF Front End Unit consists of an optional frequency converter (mixer), Power Amplifier (PA)/ Low

Noise Amplifier (LNA) and Tx/Rx filters in N Tx/Rx chains. An antenna module follows that comprises of physical antenna, RF feed distribution/aggregation network, and optionally an antenna calibration network. The timing unit includes any clock and frequency synthesis required as well as other timing and synchronization circuits. The timing unit shall derive timing and synchronization for the O-RU$_{7\text{-}2x}$ based on the IEEE 1588v2 protocol. In addition, syncE may also be used for timing as defined in O-RAN FH WG4 specifications [4].

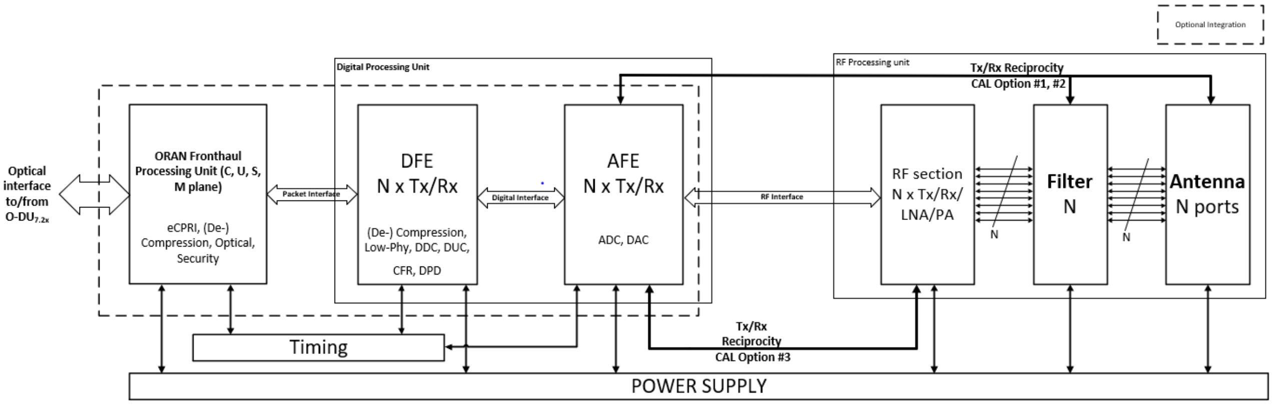

Figure 5.9-1: Generic O-RU$_{7\text{-}2x}$ Functional Module Diagram of 4T4R/8T8R

5.10 O-RU$_6$ Hardware Architecture

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5.11 O-RU$_8$ Hardware Architecture

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5.12 Integrated gNB-DU Hardware Architecture

The Integrated O-DU & O-RU (O-RU Functional Split 2) is white box hardware is the platform that performs the O- DU functions such as upper L1 and lower L2 (MAC: multiplexing, scheduling, HARQ and RLC: segmentation, re-transmission, re-ordering) functions along with O-RU. The integrated hardware consists of DU & RU in a single box as shown in Figure 5.12-1. The integrated small cell (gNB-DU) shall support split option – 2 F1 Mid-haul interface termination [3GPP] towards gNB-CU and supports FAPI standard logical interface between Layer2 and Layer1 running on different HW options.

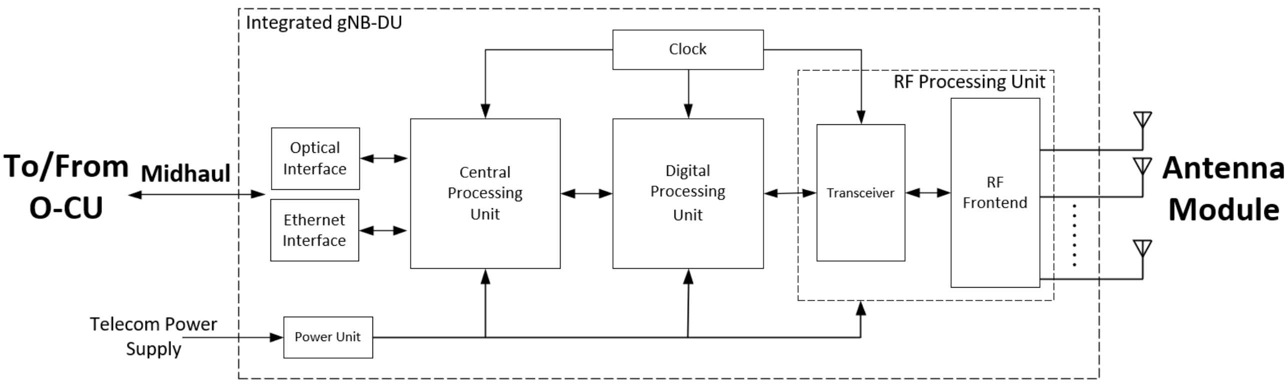

Figure 5.12-1 Integrated O-DU & O-RU (gNB-DU) O-RU Split Option 2 Reference Design

5.12.1 O-DU portion Functional Module Description

In this Integrated architecture, O- DU functions such as upper L1 and lower L2 functions. L2 Chip will be having multiple interface support like DDR4, QSPI, 1G Ethernet, SFP, PCIe, JTAG, UART, EMMC and many more to support number of additional peripherals. The O-DU portion in the integrated architecture shown in Figure 5.12-2 and it depicts the interconnection of the components and external interfaces. O-DU portion is connected to O-RU portion via PCIe interface.

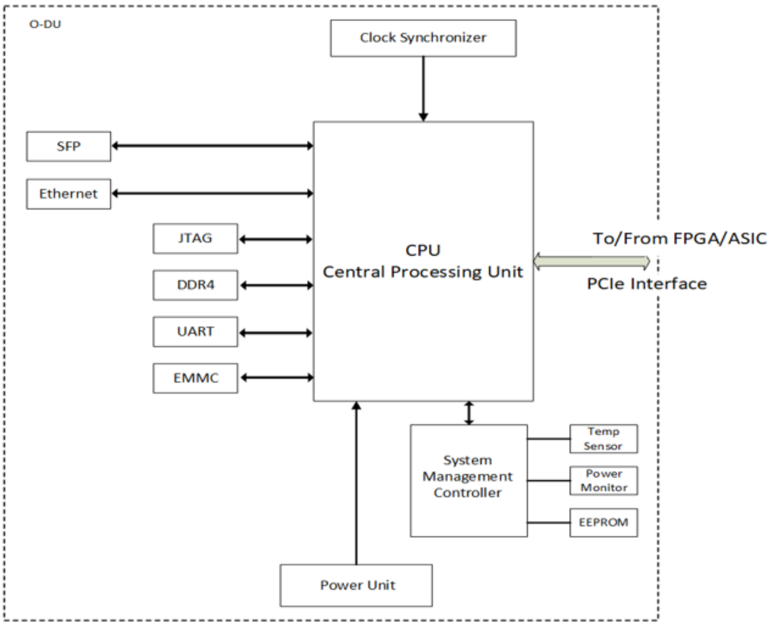

Figure 5.12-2 Functional Block Diagram for O-DU Portion

5.12.2 O-RU Portion of Integrated Reference Design

In Integrated architecture the O-RU portion contains the Digital Processing Unit Processing Unit, including the Transceiver, RFFE and other RF components, Clock and Synchronization & Power Unit. Digital processing unit is connected to O-DU portion NPU through PCIe interface.

5.12.2.1 O-RU High-Level Functional Block Diagram

Figure 5.12-3 is the high-level functional block diagram for O-RU portion in the Integrated architecture. It also highlights the internal/external interfaces that are required.

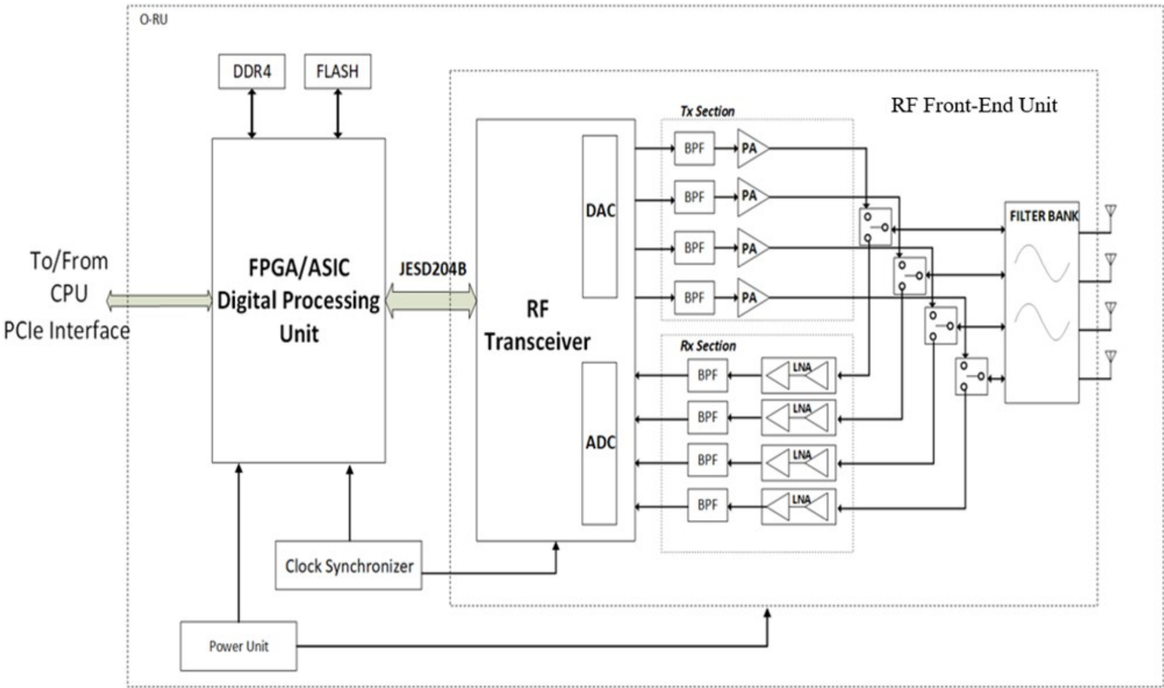

Figure 5.12-3 Functional Block Diagram for O-RU portion

5.13 mMIMO O-RU$_{7\text{-}2x}$ Functional Module Description

In this section description and diagram for three types of possible NxM mMIMO O-RU High Level System Architecture are provided. mMIMO O-RU$_{7\text{-}2x}$ can be realized with different high level radio architecture based on various functional splits and interface/interconnects between these functional blocks.

The level of integration achieved on an SoC for different functional blocks can vary and that in turn can result in different O-RU architecture and the associated interconnects between the functional blocks.

These functional splits can be realized through three possible configuration O-RU Architecture options as presented in Figures 5.13-1, 5.13-2, and 5.13-3.

In mMIMO O-RU High Level System Architecture #1 as shown in Figure 5.13-1, eCPRI frame that comes from O-DU$_{7\text{-}2x}$ is compressed/decompressed and processed by O-RAN Fronthaul Processing Unit & Beamformer block. Additionally, security compliance may be carried out according to WG11 specifications. Subsequently, Digital Processing Unit that have FPGA/ASIC based DFE module receives the packets from O-RAN Fronthaul Processing Unit & Beamformer block and performs the functionalities such as compression/decompression, Low-Phy, DDC, DUC, CFR, and DPD. The ADC and DAC functions are performed in AFE block. All these three functional blocks can be part of a single SoC or can be split into multiple physical integrated circuits (IC) and connected with digital or packet-based interface.

The functional block labelled RF Section, consists of an optional frequency converter (mixer), Power Amplifier (PA)/ Low Noise Amplifier (LNA). The filter is the next functional block in the path and is needed for each of the N Tx/Rx paths to meet the emissions in DL and provide the necessary selectivity for UL. The last functional block in the mMIMO RU system is the Antenna. Depending on the power level for each of the Tx chain in a mMIMO system, the filter could be integrated into the RF Section or could be integrated along with the antenna thus making it the AFU.

In mMIMO O-RU, Tx/Rx reciprocity calibration is one of the important steps for optimal performance. Calibration is required to align the Tx and Rx chains in phase and amplitude. In some implementations, reciprocity calibration requires to implement a dedicated feedback path. Multiple options are possible for the design of such feedback path – the main three options are shown in Figures 5.13-1, 5.13-2, and 5.13-3. The main difference between each of the calibration options is the implementation of the feedback network and where the RF feedback and associated distribution circuitry resides.

The calibration signal ultimately needs to be relayed back to the AFE, so that the phase and amplitude difference between each of the corresponding Tx-Rx chain and the phase and amplitude differences between the multiple Tx-Rx chains can be estimated and calibrated accurately to ensure uplink and downlink channel reciprocity to enable beamforming. In CAL option #1, the RF feedback and the distribution circuitry resides in the antenna module and all the RF components including the antenna are part of the Tx/Rx reciprocity calibration loop. In CAL option #2, the RF feedback and the distribution circuitry resides in the filter module and all the RF components till the filter module are part of the Tx/Rx reciprocity calibration. In CAL option #3, the RF feedback and the distribution circuitry resides in the RF Section and all the RF components till the RF Section's output in the Tx path and input in the Rx path are part of the Tx/Rx reciprocity calibration. These calibration options (CAL option #1, #2 and #3) can apply to all three mMIMO O-RU architectures in Figures 5.13-1, 5.13-2, and 5.13-3.

The timing unit includes any clock and frequency synthesis required as well as other timing and synchronization circuits. The Timing Unit shall derive timing and synchronization for the O-RU$_{7\text{-}2x}$ based on the IEEE 1588v2 protocol. In addition, syncE may also be used for timing as defined in O-RAN FH WG4 specifications [4].

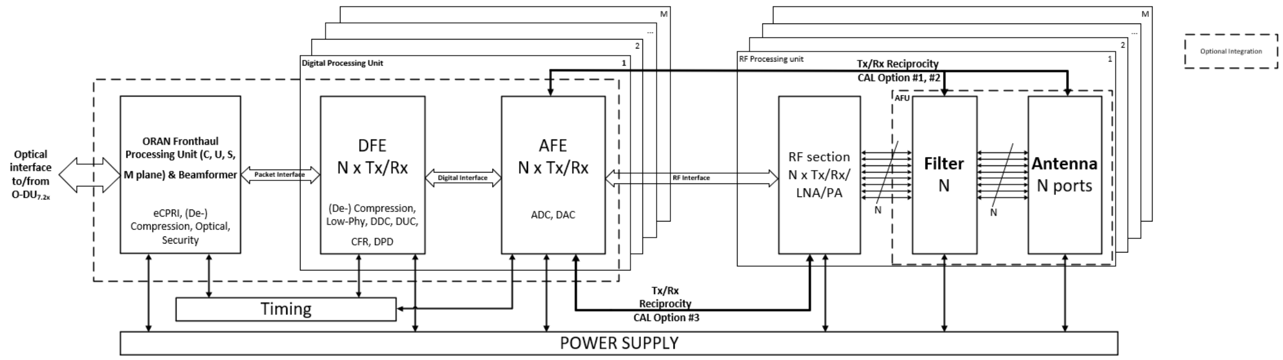

Figure 5.13-1: mMIMO [NxM]T[NxM]R O-RU High Level System Architecture #1

In mMIMO O-RU High Level System Architecture #2, the operations that are performed in DFE of mMIMO O-RU High Level System Architecture #1 are distributed between the O-RAN Fronthaul Processing Unit & Beamformer and AFE as shown in Figure 5.13-2. In particular, the Low-Phy functions of DFE are performed in the ASIC/FPGA for O-RAN Fronthaul Processing Unit & Beamformer and DDC, DUC, CFR, and DPD blocks of DFE are part of AFE in the RFIC as shown in Figure 5.13-2. The interconnect between the ASIC/FPGA for ORAN Fronthaul Processing Unit & Beamformer.

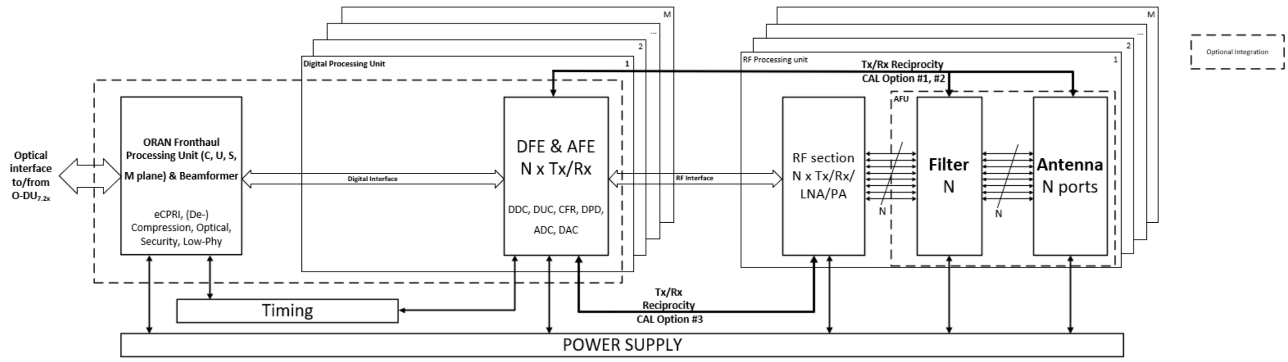

Figure 5.13-2: mMIMO [NxM]T[NxM]R O-RU High Level System Architecture #2

In mMIMO O-RU High Level System Architecture #3, DFE and AFE functions are combined on a single DFE & AFE functional module as shown in Figure 5.13-3. In this architecture, (De-)Compression, Low-Phy, DDC, DUC, CFR, and DPD are part of same FPGA/ASIC. There is a packet-based interface/interconnect between the O-RAN Fronthaul Processing Unit & Beamformer and the DFE & AFE functional module. These modules can be on a single SoC or can be on different FPGA/ASIC.

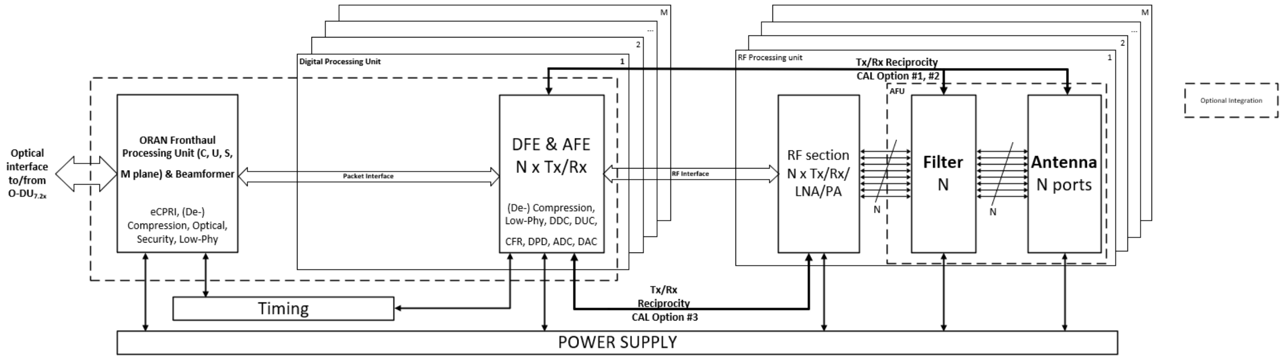

Figure 5.13-3: mMIMO [NxM]T[NxM]R O-RU High Level System Architecture #3

6 White Box Hardware Requirements

This chapter provides the requirements for various white boxes used within the outdoor macro cell base station. These white boxes are O-CU, O-DU$_{7-2x}$ and O-RU$_{7-2x}$. The O-CU and O-DU$_{7-2x}$ can be implemented in an integrated fashion into one white box hardware or they can be separated. The details of the testing and compliance to these requirements would be taken as part of the TIFG specification [11]. These requirements will be used to define a hardware reference design specification for the outdoor macro cell base station [12].

6.1 O-CU Requirements

O-CU requirements are described in the following sections, which include the performance, interface, environmental, EMC, mechanical, thermal and power requirements.

6.1.1 O-CU Performance

The performance requirements of the O-CU are listed in Table 6.1-1.

Table 6.1-1 :O-CU Performance Requirements

Parameter	Requirement	Priority
Synchronization	• The solution shall support 1588v2 Synchronization. • The solution shall support T-TSC (Telecom Time Slave Clock).	High
Peak Data Rate	Peak data rate shall be calculated as defined in 3GPP TS 38.306	
Latency	Control Plane < 20 ms (def: message1 to message 5) User Plane < 4 ms (def: PDCP SDU → PDCP SDU)	High

6.1.2 O-CU Interfaces

The interface requirements of the O-CU are listed in Table 6.1-2

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Table 6.1-2: O-CU Interface Requirements

Parameter	Requirement	Priority
Transport Interface	1/10 GbE F1 Interface to connect with O-DU$_{7\text{-}2x}$ at cell site or 25 GbE F1 Interface to connect with O-DU$_{7\text{-}2x}$ at data center	High
	1/10 GbE NG interface to connect with 5G core when O-DU$_{7\text{-}2x}$ is at cell site or 25/100 GbE NG interface to connect with 5G core when O-DU$_{7\text{-}2x}$ is at data center	High

6.1.3 O-CU Environmental and EMC

The EMC requirements of the O-CU are listed in Table 6.1-3.

Table 6.1-3: O-CU EMC Requirements

Parameter	Requirement	Priority
Safety, EMC	The solution shall comply with the requirements of 3GPP TS 38.113 (2017-12R15) [7] for equipment used in telecommunication room	High
Safety, EMC	The solution shall be certified for the required regulatory compliance related to EMC and safety	High
Safety, Environment and Reliability	The solution shall support Front to Back airflow	High
Safety, Environment and Reliability	The solution shall support Front Serviceability	High
Safety, Environment and Reliability	The solution should support IP65 rated version (if implemented outdoors).	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment [17]	High

6.1.4 O-CU Mechanical, Thermal and Power

The mechanical, thermal, and power requirements of the O-CU are listed in Table 6.1-4 and Table 6.1-5.

Table 6.1-4: O-CU Mechanical Requirements

Parameter	Requirement	Priority
Dimension	The solution shall be 1 rack unit in height	High
Dimension	The sled shall meet dimensions of 41 mm (Height) x 215 mm (Width) x 427.5 mm (Depth)	High
Status Indicator LED	At least include the following status indicators: • 1 LED indicating on/off status of the power supply • 1 LED indicating on/off status of the transmission link	High

Power Supply	The solution shall support -48V DC power. The tolerance range should be within -42V and -58V DC	High
Power Dissipation	The maximum power consumption of the solution in normal conditions shall not exceed 400W	High
Power Drain	The solution shall have the ability to reduce the power drain when not at full load	High

Table 6.1-5: O-CU Thermal Requirements

Parameter	Requirement	Priority
Environment and Reliability	The solution shall support operating temperature ranges from 0℃ to 50℃ indoors.	High
Environment and Reliability	The solution may have FANs to support temperature requirements.	High
Environment and Reliability	The solution should support extended temperature range (-40℃ to +65 ℃) if implemented outdoors.	Low

6.2 O-DU$_x$ Common Requirements

The common requirements apply to all the O-DU$_X$ regardless of the split option. However, in this version of the specification, the focus will be on O-DU$_{7-2x}$. If more options get standardized in a future version of this document, some of these common requirements may be moved to the specific sub-sections in section 6.3.

6.2.1 O-DU$_{7-2x}$ Performance

The performance requirements of the O-DU$_{7-2x}$ are listed in Table 6.2-1.

Table 6.2-1: O-DU$_{7-2x}$ Performance Requirements

Parameter	Requirement	Priority
O-DU$_{7-2x}$ server Processing	For the O-DU$_{7-2x}$ at cell site serving the FDD bands with 4T4R radios, the solution shall support at least 210 MHz in downlink and 105 MHz in uplink For the O-DU$_{7-2x}$ at data center serving the FDD bands with 4T4R radios, the solution shall support at least 2 x 210 MHz in downlink and 2 x 105 MHz in uplink	High
Transmission distance	Directly connected with O-RUs at cell site or Connected with O-RUs 40 km apart	High

Synchronization	For O-DU$_{7-2x}$ at cell site:
	• The solution shall have a built-in GPS receiver for timing that supports PTP distribution compliant to G.8272.
	• The solution shall comply with the following ITU-T specifications for SyncE:
	  o Definitions: ITU-T G.8260
	  o Architecture: ITU-T G.8261	

	o SSM transport channel and format: ITU-T G.8264 o Clock specifications: ITU-T G.8262 (EEC) o Functional model and SSM processing: ITU-T G.781 • The solution shall support IEEE 1588 Telecom Grand Master (T-TGM) using G8275.1 profile. • The solution shall support T-GM conversion to Telecom Boundary Clock (T-TBC) when GPS is lost. • The solution shall support Telecom Boundary Clock (T-TBC) per ITU-T G.8273.2. • The solution shall meet 4 hour holdover time of frequency, time/phase. • The solution shall comply with input and output jitter and wander requirements specified in ITU-T G.8262 (for EEC). • For timing synchronization, the solution shall support ORAN LLS-C1/C2 configuration. • The solution shall support alternate BMCA specified in G.8275.1	
Synchronization	For O-DU$_{7-2x}$ at data center: • For timing synchronization, the solution shall support ORAN LLS- C3 configuration. • The solution shall support T-TSC.	High
Peak Rate	Peak data rate to be calculated based on the formula provided in 3GPP TS 38.306	High
Modulation	DL: QPSK, 16QAM, 64QAM, 256QAM, (1024 QAM desired) UL: π/2-BPSK, QPSK, 16QAM, 64QAM (256QAM/1024 QAM desired)	High
Latency	Up to 200 us fronthaul latency	High

6.2.2 O-DU$_{7-2x}$ Interfaces

The interface requirements of the O-DU$_{7-2x}$ are listed in Table 6.2-.

Table 6.2-2: O-DU$_{7-2x}$ Interface Requirements

Parameter	Requirement	Priority
Transport Interface	Fronthaul interfaces connected with either radio unit or fronthaul gateway	High
	At least one NG interface to 5GC	High
	At least one 1/10 GbE F1 Interface to connect with O-DU$_{7-2x}$ at cell site or 25 GbE F1 Interface to connect with O-DU$_{7-2x}$ at data center	High

6.2.3 O-DU$_x$ Environmental and EMC

The EMC requirements of the O-DU$_{7-2x}$ are listed in Table 6.2-3.

Table 6.2-3: O-DU$_{7-2x}$ EMC Requirements

Parameter	Requirement	Priority
Safety, EMC	The solution shall comply with the requirements of 3GPP TS 38.113 (2017-12R15) [7] for equipment used in telecommunication room	High
Safety, EMC	The solution shall be certified for the required regulatory compliance related to EMC and safety	High

Safety, Environment and Reliability	The solution shall support Front to Back airflow	High
Safety, Environment and Reliability	The solution shall support Front Serviceability	High
Safety, Environment and Reliability	The solution should support IP65 rated version. (if implemented outdoors)	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment [17]	High

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6.2.4 O-DU$_x$ Mechanical, Thermal and Power

The mechanical and power requirements of the O-DU$_{7-2x}$ are listed in Table 6.2-4.

Table 6.2-4: O-DU$_{7-2x}$ Mechanical Requirements

Parameter	Requirement	Priority
Dimension	The solution shall be 1 rack unit in height	High
Dimension	For the O-DU$_{7-2x}$ at cell site, the solution dimensions shall not exceed 19 inch width and 18 inch depth For the O-DU$_{7-2x}$ at data center, the sled shall meet dimensions of 41 mm (Height) x 215 mm (Width) x 427.5 mm (Depth)	High
Dimension	The solution shall have PCIe Gen 3/4 x16 low profile or half-length slot to add to support FEC Accelerator	High
Dimension	The solution shall support a minimum of 64 GB DDR4 or better DRAM using all available DIMM slots to maximize memory bandwidth. 8 DIMM slots are preferred.	High
Dimension	The solution shall have at least 2x 240GB SSD drives	High
Dimension	For the O-DU$_{7-2x}$ at the cell site, the solution with minimum 16 Core and clock speed of 2.3 GHz is recommended. Equivalent/higher alternative processor technology and core counts including combinations of CPUs, GPUs, FPGAs and ASICs and clock frequencies are acceptable. For the O-DU$_{7-2x}$ at the data center, the solution with minimum 24 Core and clock speed of 2.4 GHz is recommended. Equivalent/higher alternative processor technology and core counts including combinations of CPUs, GPUs, FPGAs and ASICs and clock frequencies are acceptable.	High
Dimension	For the O-DU$_{7-2x}$ at the cell site, the solution shall have at least 6x 10G ports to carry eCPRI ORAN 7-2x fronthaul traffic from O-RUs to the O-DU$_{7-2x}$ For the O-DU$_{7-2x}$ at the data center, the solution shall have at least 4x 25G ports to carry eCPRI fronthaul and midhaul traffic	High
Dimension	For the O-DU$_{7-2x}$ at the cell site, the solution shall have 2x 1/10G ports to carry ORAN Split 2 midhaul traffic from O-DU$_{7-2x}$ to O-CU	High
Status Indicator LED	At least include the following status indicators: • 1 LED indicating on/off status of the power supply • 1 LED indicating on/off status of the transmission link	High

Power Supply	For the O-DU$_{7\text{-}2x}$ at the cell site, the solution shall have a single DC power supply. For the O-DU$_{7\text{-}2x}$ at the data center, the solution shall have a single AC/DC power supply. The solution shall support -48V DC power. The tolerance range should be within -42V and -58V DC	High
Power Dissipation	The maximum power consumption of the solution in normal conditions shall not exceed 400W	High
Power Drain	The solution shall have the ability to reduce the power drain when not at full load	High

The thermal requirements of the O-DU$_{7\text{-}2x}$ are listed in Table 6.2-5.

Table 6.2-5: O-DU$_{7\text{-}2x}$ Thermal Requirements

Parameter	Requirement	Priority
Environment and Reliability	The solution shall support operating temperature ranges from 0℃ to 50℃ for data center.	High
Environment and Reliability	The solution may have FANs to support temperature requirements.	High
Environment and Reliability	The solution should support extended temperature range (-40℃ to +55℃) if implemented outdoors.	Low
Environment and Reliability	The solution shall support in-band Redfish Chassis management	High
Environment and Reliability	The solution shall have at least 4 dry contact inputs and ability for environmental and cabinet alarms monitoring.	High

6.3 O-DU$_x$ Split Option Specific Requirements

6.3.1 O-DU$_{7\text{-}2x}$ Specific Requirements

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6.3.2 O-DU$_6$ Specific Requirements

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6.3.3 O-DU$_8$ Specific Requirements

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6.4 O-RU$_x$ Common Requirements

The common requirements apply to all the O-RU$_X$ regardless of the split option. However, in this version of the specification, the focus will be on O-RU$_{7\text{-}2x}$. If more options get standardized in a future version of this document, some of these common requirements may be moved to the specific sub-sections in section 6.5.

In the outdoor environment, the O-RU$_{7-2x}$ hardware is placed on the rooftop, pole or wall of the coverage building; it converts the baseband signals to RF signals for transmission over the air or vice versa. All radio requirements are subject to local regulatory limits, where applicable.

6.4.1 O-RU$_{7-2x}$ Performance

The O-RU$_{7-2x}$ performance requirements cover all the aspects of radio unit including frequency bands, antenna configurations, power efficiency, etc. Table 6.4-1 lists the performance parameters related to O-RU$_{7-2x}$.

Table 6.4-1: O-RU$_{7-2x}$ Performance Requirements

Parameter	Requirement	Priority
Operating band	B1/n1, B3/n3, B7/n7, n26, n29, n66, n70, n71, n77, n78, n78 European sub band (3400-3800 MHz according with ECC Decision (11)06 and CEPT Report 67). All the bands according to 3GPP TS 36.104 [13] and 3GPP TS 38.104 [3] except n78 European sub-band.	High
Bandwidth	The IBW of the n66 O-RU$_{7-2x}$ shall be 90 MHz The OBW of the n66 O-RU$_{7-2x}$ shall be 90MHz.	High
	The DL IBW of the n70 O-RU$_{7-2x}$ shall be 25MHz, and its UL IBW shall be 15MHz. The DL OBW of the n70 O-RU$_{7-2x}$ shall be 25MHz, and its UL OBW shall be 15MHz.	
	The IBW of the n26 O-RU$_{7-2x}$ shall be 7MHz. The OBW of the n26 O-RU$_{7-2x}$ shall be 7MHz.	
	The IBW of the n29 O-RU$_{7-2x}$ shall be 11MHz. The OBW of the n29 O-RU$_{7-2x}$ shall be 11MHz.	
	The IBW of the n71 O-RU$_{7-2x}$ shall be 35MHz. The OBW of the n71 O-RU$_{7-2x}$ shall be 35MHz.	
	The IBW of the n77 O-RU$_{7-2x}$ shall be 280MHz. The OBW of the n77 O-RU$_{7-2x}$ shall be 200MHz.	
	The IBW of n78 O-RU$_{7-2x}$ shall be 400MHz. The OBW of n78 O-RU$_{7-2x}$ shall be up to 300MHz.	
	The IBW of B1/n1 O-RU$_{7-2x}$ shall be 60MHz. The OBW of B1/n1 O-RU$_{7-2x}$ shall be 60MHz.	
	The IBW of B3/n3 O-RU$_{7-2x}$ shall be 75MHz. The OBW of B3/n3 O-RU$_{7-2x}$ shall be 75MHz.	
	The IBW of B7/n7 O-RU$_{7-2x}$ shall be 70MHz. The OBW of B7/n7 O-RU$_{7-2x}$ shall be 70MHz.	
O-RU$_{7-2x}$ Multi-band operation	Tri-band: n26+n29+n71 The O-RU$_{7-2x}$ shall have at least 4 RF ports, with n71, n29, and n26 sharing the same RF ports. Tri-band: B1/n1+B3/n3+B7/n7	High

	The O-RU$_{7-2x}$ shall have at least 4 RF ports, with B1/n1, B3/n3, and B7/n7 sharing the same RF ports.	
	Dual-band: n66+n70 The O-RU$_{7-2x}$ shall have at least 4 RF ports, with n66 and n70 sharing the same RF ports. Dual-band: B1/n1+B3/n3 The O-RU$_{7-2x}$ shall have at least 4 RF ports, with B1/n1 and B3/n3 sharing the same RF ports. Dual-band: B1/n1+B7/n7 The O-RU$_{7-2x}$ shall have at least 4 RF ports, with B1/n1 and B7/n7 sharing the same RF ports.	High
Expected Multi-mode and Transceiver Configuration (Number of Transmit/Receive Paths)	Tri-band (n26+n29+n71): 4T4R Tri-band (B1/n1+B3/n3+B7/n7): 4T4R	High
	Dual-band (n66+n70): 4T4R Dual-band (B1/n1+B3/n3 or B3/n3+B7/n7): 4T4R	High
	Single-band (n77): 8T8R	High
	Single-band (n77): 64T64R	High
	Single-band (n78): 32T32R and 64T64R	High
	Single-band (B1/n1 or B3/n3 or B7/n7): 4T4R	High
Output Power Accuracy	The EIRP/output power of the O-RU$_{7-2x}$ shall follow 3GPP TS38.141-1 and TS38-141-2 requirements for normal and extreme conditions as well as for radiated and conducted tests. (It is desirable to remain within +/- 1dB of the set power under all operation condition.)	High
TX off Power Level	Less than -85 dBm/MHz (-89 dBm/MHz desired) per antenna connector or TAB connector as per 3GPP TS 38.104 [3]	High
EVM at maximum output power	64QAM: EVM smaller than 8% (5% desired) 256QAM: EVM smaller than 3.5% For other modulations, refer to 3GPP TS 38.104 [3]	High
Operating band unwanted emissions	The operating band unwanted emissions must satisfy the Category B limit defined by the section 6.6.4 in 3GPP TS 38.104 [3].	High
Transmitter spurious emissions	The Operating band unwanted emissions must satisfy the Category B limit defined by the section 6.6.5 in 3GPP TS 38.104 [3].	High
Noise Figure	O-RU$_{7-2x}$ receiver Uplink Noise Figure shall be better than 1.9dB typical (average value in normal conditions), 2.5dB Max (worst case conditions) for single-band, dual-band and tri-band. For massive MIMO, the noise figure requirement shall be lower than 3.5 dB	High
Rx sensitivity	The throughput shall be ≥ 95% of the maximum throughput of the reference measurement channel as specified in annex A.1 with parameters specified in 3GPP TS 38.104 table 7.2.2-1 for Wide Area BS.	High

Blocking	In channel selection, ACS, In-band blocking, out-band blocking, IMD and other receiver specification must follow the 3GPP guidelines for the reference sensitivity is allowed to degrade at most 6 dB under all scenarios of interference signal and corresponding level, as per table 7.4.1.2-1 of [3].	High
Other specifications	Except for all the RF specifications listed above, other RF specifications must follow the requirement in [3].	High
Modulation Mode	DL: QPSK, 16QAM, 64QAM, 256QAM, (1024 QAM desired) UL: π/2-BPSK, QPSK, 16QAM, 64QAM (256QAM/1024 QAM desired)	High
Min. Output power	The tri-band O-RU$_{7\text{-}2x}$ (n26+n29+n71):	High
	4T4R - n26 shall be 10W per port 4T4R - n29 shall be 40W per port 4T4R - n71 shall be 30W per port The tri-band O-RU$_{7\text{-}2x}$ (B1/n1+B3/n3+B7/n7): 4T4R – B1/n1 shall be 40W per port 4T4R – B3/n3 shall be 40W per port 4T4R – B7/n7 shall be 40W per port	High
	The dual-band O-RU$_{7\text{-}2x}$ (n66+n70): The combined output power shall be at least 80W per port, under all operating conditions. The O-RU$_{7\text{-}2x}$ shall be able to allocate higher power level for n66 (example: 60W) and lower power level for n70 (example: 20W). The power setting shall be configurable by the operator. The dual-band O-RU$_{7\text{-}2x}$ (B1/n1+B3/n3 or B1/n1+B7/n7): The combined output power shall be at least 80W per port, under all operating conditions. The O-RU$_{7\text{-}2x}$ shall support equal power allocation between bands and preferably should be able to allocate higher power level for B1/n1 (example: 60W) and lower power level for B3/n3 or B7/n3 (example: 20W) and vice versa. The power setting shall be configurable by the operator.	High
	Single band O-RU$_{7\text{-}2x}$ (n77): - 8T8R: 40W per port - 64T64R: 80dBm EIRP (integrated antenna) Single band O-RU$_{7\text{-}2x}$ (B1/n1 or B3/n3): - 4T4R: 40W per port Single band O-RU$_{7\text{-}2x}$ (B7/n7): - 4T4R: 60W per port	High
	For any Wide band O-RU$_{7\text{-}2x}$ (n78) product (32T32R and 64T64R) the EIRP shall be (for narrow beam at boresight; assuming SU-MIMO; no power sharing between beams): > +78 dBm (Typically 55 dBm output power + 23 dBi)	High
Regulation	For n26 band, the O-RU$_{7\text{-}2x}$ shall have a feature limiting its TX power and meet FCC rules on output power as outlined in part 90 and part 22.	High
EIRP Stability	The EIRP/output power of the O-RU$_{7\text{-}2x}$ shall follow 3GPP TS38.141-1 and TS38-141-2 requirements for normal and extreme conditions as well as for radiated and conducted tests. (It is desirable to remain within +/- 1dB of the set power under all operation condition.)	High

Synchronization	The O-RU$_{7-2x}$ shall support GPS and/or IEEE 1588v2 for timing synchronization based on O-RAN CUS plane specification [4].	High
	The O-RU$_{7-2x}$ shall support Telecom Slave Clock (T-TSC) to the protocol(s) as defined for the fronthaul interface. Furthermore, the O-RU$_{7-2x}$ shall provide additional noise filtering to filter fronthaul interface dynamic noise which will help meeting 3GPP frequency accuracy requirement	High
	For timing synchronization, the O-RU$_{7-2x}$ shall support the mandatory configurations as defined by O-RAN WG4 specification [4]	High
Optical Interface	For 4T4R radios: - For n66, n70, n26, n29, n71 radios: At least one SFP 10G interface. - For B1/n1, B3/n3 and B7/n7: At least 2x10Gbps SFP+.	High
	For any M-MIMO products the minimum number of Front haul Interface PHY ports should be 2x25Gpps, with SFP28 optical connectors, although 4x25Gbps SFP28 should be needed depending on TRx configuration, IBW, OccBW, total number of carriers supported; DL/UL MU_MIMO paired layers supported, etc.	High
Beamforming support (for massive MIMO radios)	Number of Beams: Up to 8 SSB beams (FR1)	High
	32T32R: Horizontal plane: at least ±45 degrees (±60 degrees desirable) with side lobes suppression > 10dB (>15 dB desirable). 64T64R: Horizontal plane: at least ±45 degrees (±60 degrees desirable) with side lobes suppression > 10dB (>15 dB desirable).	High
	32T32R: Vertical plane: At least ±5 degrees with side lobes suppression > 6dB (>10 dB desirable) At least ±2 degrees with side lobes suppression > 10dB (>15 dB desirable) 64T64R: Vertical plane: At least ±15 degrees with side lobes suppression > 6dB (>10 dB desirable) At least ±10 degrees with side lobes suppression > 10dB (>15 dB desirable)	High
Antenna elements	For M-MIMO products n78 or any subsets inside this band, the minimum number of single polarization antenna elements must be 192: (12 rows x8 columns x 2 dual slant cross polar antenna elements).	High
AAU Number of simultaneous TX user beams/layers per carrier	16DL and 8UL	High

6.4.2 O-RU$_{7-2x}$ Interfaces

The interface related requirements of the O-RU$_{7-2x}$ are listed in Table 6.4-2.

Table 6.4-2: O-RU$_{7-2x}$ Interface Requirements

Parameter	Requirement	Priority

Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall be compliant to O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall be compliant to O-RAN M-plane specification [10]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall comply to M-Plane IOT Profile 1 specified in O-RAN M-plane specification [10]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support at least one of the following NETCONF/YANG based models for management plane as specified in O-RAN M-plane specification [10]: (a) hierarchical architecture model (b) hybrid architecture model	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support direct connection between O-RU$_{7\text{-}2x}$ and O-DU$_{7\text{-}2x}$ for fronthaul synchronization as specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support LLS-C2 and LLS-C3 bridged connection between O-RU$_{7\text{-}2x}$ and O-DU$_{7\text{-}2x}$ for fronthaul synchronization as specified in O-RAN CUS plane specification [4].	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall comply to all the test cases specified in O-RAN IOT specification [11]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support Block floating point compression as specified in O-RAN CUS plane specification [4]. The compression bits shall be configurable through EMS.	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support Pre-configured Transport Delay Method as specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support transmission blanking as specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support the counters specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support the eCPRI header format, section type and section extension type messages as specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support G.8275.1 synchronization as specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support G.8275.2 synchronization as specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall support IEEE 1914.3, ITU-T G.826x, G.827x series specified in O-RAN CUS plane specification [4]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall be compliant to security requirements specifications in accordance with WG11 technical specifications [14]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall be compliant to security protocol specifications in accordance with WG11 technical specifications [15]	High
Fronthaul interface	The O-RU$_{7\text{-}2x}$ shall be compliant to security test specifications in accordance with WG11 technical specifications [16]	High

6.4.3 O-RU$_{7\text{-}2x}$ Environmental and EMC

The environmental and EMC requirements of the O-RU$_{7\text{-}2x}$ are listed in Table 6.4-3.

Table 6.4-3: O-RU$_{7\text{-}2x}$ Environmental and EMC Requirements

Parameter	Requirement	Priority
Mounting method	Back of antenna, Tower, Roof top, Pole, or Wall	High
RF Connector	RF connector type should be 4.3-10 female. Location should be on the bottom of O-RU$_{7-2x}$, on left side of connector panel arranged in a single 1x4 row and dimensioned from the backside face.	High
AISG Connector	AISG Connector should be standard 8 pin female connector. Location should be on the bottom of O-RU$_{7-2x}$ and dimensioned from the backside face.	High
RET Connector	O-RU$_{7-2x}$ shall provide direct support of antenna Remote Electrical Tilt via a specific port for RET: DIN-8 RET, AISG 2.0. This port shall be located down of the unit.	High
Optical Connector	Optical Connector should be SFP+/SFP28. Location should be on the bottom of the O-RU$_{7-2x}$ and dimensioned from the backside face.	High
Power Connector	Power connector should be: DC Circular Connector, Receptacle, 2-Pin Male. Location should be on the bottom of O-RU$_{7-2x}$ and dimensioned from the backside face. Also, connector should be located on the far-right side of connector panel to allow ease of installation	High
Lightning Protection	O-RU$_{7-2x}$ shall have lightning protection	High
Grounding	The O-RU$_{7-2x}$ shall support grounding. Ground connector should be a two-wire ground lug. Grounding lug location should be on the bottom of the O-RU$_{7-2x}$ and dimensioned from the backside face	High
EMC, Safety, Storage, and Transportation	The O-RU$_{7-2x}$ shall comply to the following EMC, Safety, Storage, and Transportation Requirements: CFR 47, Part 1.130, Part 15B, UL 62368-1, UL 50E, UL 60950-22, 3GPP 38.113, ETSI EN 300 019-2-4, Severity class 1, Telcordia GR-63, Zone4, Telcordia GR 487, Telcordia GR 63, ETSI 300 019-1-1 Class 1.3, and IEC-61000-4-5, ETSI EN 301 489-1, 3GPP TS 37.113, 3GPP TS 37.114, Directive 2002/95/EC & Directive 2011/65/EU, Directive 2002/96/EC, EU Radio Equipment Directive (RED) 2014/53/EU, ETSI EN 300 019-1-2, Class 2.3, ETSI EN300019-1-1 V2.1.4 (2003-04) Class 1.2: "Weather protected, not temperature-controlled storage locations", ETSI EN 300 019-1-3, class 3.2 (indoor site), ETSI EN 300 019-1-4, class 4.1 (outdoor site).	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment [17]	High

6.4.4 O-RU$_{7-2x}$ Mechanical, Thermal and Power

The mechanical, thermal and power requirements of the O-RU$_{7-2x}$ are listed in Table 6.4-4.

Table 6.4-4: O-RU$_{7-2x}$ Mechanical, Thermal and Power Requirements

Parameter	Requirement	Priority
Weight (excluding mounting kit)	The weight of the tri-band O-RU$_{7-2x}$ (n26+n29+n71) shall be less than 35kg.	High
	The weight of the dual-band O-RU$_{7-2x}$ (n66+n70) shall be less than 30kg.	High
	The weight of the single band 64T64R O-RU$_{7-2x}$ (n77) shall be less than 50kg	High
	The weight of the single band 8T8R O-RU$_{7-2x}$ (n77) shall be less than 33kg	High
	The weight of the single band 32T32R O-RU$_{7-2x}$ (n78) shall be less than 30kg The weight of the single band 64T64R O-RU$_{7-2x}$ (n78) shall be less than 40 kg	High

	The weight of the single band 4T4R O-RU$_{7-2x}$ (B1/n1 or B3/n3) shall be less than 16kg	High
	The weight of the single band 4T4R O-RU$_{7-2x}$ (B7/n7) shall be less than 18kg	High
	The weight of the dual band 4T4R O-RU$_{7-2x}$ (B1/n1+B3/n3 or B3/n3+B7/n7) shall be less than 33kg	High
	The weight of the tri-band 4T4R O-RU$_{7-2x}$ (B1/n1+B3/n3+B7/n7) shall be less than 35kg	High
Volume	The volume of the tri-band O-RU$_{7-2x}$ (n26+n29+n71) shall be less than 33 Liter.	High
	The volume of the dual-band O-RU$_{7-2x}$ (n66+n70) shall be less than 28 Liter.	High
	The volume of the single-band 64T64R O-RU$_{7-2x}$ (n77) shall be less than 81 Liters.	High
	The volume of the single-band 8T8R O-RU$_{7-2x}$ (n77) shall be less than 42.5 Liters.	High
	The volume of the single-band 32T32R O-RU$_{7-2x}$ (n78) shall be less than 45 Liters. The volume of the single-band 64T64R O-RU$_{7-2x}$ (n78) shall be less than 55 Liters.	High
	The volume of the single-band 4T4R O-RU$_{7-2x}$ (B1/n1, B3/n3) shall be less than 16 Liters.	High
	The volume of the single-band 4T4R O-RU$_{7-2x}$ (B7/n7) shall be less than 18 Liters.	High
	The volume of the dual-band 4T4R O-RU$_{7-2x}$ (B1/n1+B3/n3 or B3/n3+B7/n7) shall be less than 24 Liters.	High
	The volume of the tri-band 4T4R O-RU$_{7-2x}$ (B1/n1+B3/n3+B7/n7) shall be less than 34 Liters.	High
Dimensions	Maximum dimensions of tri-band O-RU$_{7-2x}$ (n26+n29+n71) and dual-band O-RU$_{7-2x}$ (n66+n70) should be 462mm (L) x 340mm (W) x 177 mm (D). Physical dimensions shall not exceed 890mm x 500mm x 250mm for 64T64R n77 radio. Physical dimensions shall not exceed 420mm x 420mm x 240mm for 8T8R n77 radio. For 64T64R n78 radio L < 800mm x W < 400mm x D < 250 mm. (including connectors and radome/sunshield).	High
Mounting Holes	A pattern of M8 mounting holes should be located on backside of the O-RU$_{7-2x}$, with a total of 8 holes (4 towards top and 4 towards bottom).	High
Reliability	MTBF ≥ 250.000 hours for mMIMO MTBF ≥ 300.000 hours for non mMIMO; In normal operating conditions (ETSI 300 019 environment temperature +25ºC).	High
Power Consumption	The tri-band O-RU$_{7-2x}$ (n26+n29+n71): shall be less than 1,200 Watt at 100% full loading, under all operational conditions.	High
	The dual-band O-RU$_{7-2x}$ (n66+n70): shall be less than 1,200W at 100% full loading, under all operational conditions.	High
	The single-band 64T64R O-RU$_{7-2x}$ (n77) shall be less than 1,810W at 100% full loading, under all operational conditions. Typical power measurement for 100% load may be done following 3GPP procedures for TDD with 7D1S2U configuration using TM1.1 with ~75% DL ratio in the slot.	High
	The single-band 32T32R O-RU$_{7-2x}$ (n78) shall be less than 800W at 100% full loading, under all operational conditions. Typical power measurement for 100% load may be done following 3GPP procedures for TDD with 7D1S2U configuration using TM1.1 with ~75% DL ratio in the slot.	High

	The Wide band 64T64R O-RU$_{7-2x}$ (n78) shall be less than 1000W, assuming at 100% full loading, under all operational conditions and 55ºC, 320W output power. Typical power measurement for 100% load may be done following 3GPP procedures for TDD with (DDDSU, with special subframe 10:2:2 slot format) configuration using TM1.1 with ~80% DL ratio in the slot. No fans.	High
	For O-RU$_{7-2x}$ Single band B1/n1 or Single band B3/n3 products shall be less than 500W, assuming at 100% full loading, under all operational conditions. At 55ºC and for natural cooling (no fans).	High
	For O-RU$_{7-2x}$ Single band B7/n7 products shall be less than 700W, assuming at 100% full loading, under all operational conditions. At 55ºC and for natural cooling (no fans).	High
	For O-RU$_{7-2x}$ Dual band B1/n1+B3/n3 or B3/n3+B7/n7 products shall be less than 950W, assuming at 100% full loading, under all operational conditions. At 55ºC and for natural cooling (no fans).	High
	For O-RU$_{7-2x}$ Tri-band B1/n1+B3/n3+B7/n7 products shall be less than 1300W, assuming at 100% full loading, under all operational conditions. At 55ºC and for natural cooling (no fans).	High
Power supply	The O-RU$_{7-2x}$ shall operate with the nominal voltage of -48V	High
	The O-RU$_{7-2x}$ shall have protection from power transients and surges with no impact/degradation to its performance or operation.	High
	In case of power loss to the O-RU$_{7-2x}$, and after the power is restored, the O-RU$_{7-2x}$ shall be fully functional and operational without any human interaction/intervention.	High
Level of protection	O-RU$_{7-2x}$ shall be at a minimum IP65 rated if deployed outdoors	High
Temperature	O-RU$_{7-2x}$ shall operate at the temperature range of -40℃ to ＋55℃ with both cold start and hot start options.	High
Humidity	O-RU$_{7-2x}$ shall operate between 0.03 g/m^3 and 30 g/m^3 of absolute humidity	High
Atmospheric pressure	O-RU$_{7-2x}$ shall operate between 70 kPa and 106 kPa atmospheric pressure	High
Cooling mode	O-RU$_{7-2x}$ shall support convection air cooling	High

6.5 O-RU$_{7-2x}$ Split Option Specific Requirements

Besides the common requirements which shall apply to all the radio unit types. The following sections list all the specific requirements that only apply to the designated split option.

6.5.1 O-RU$_{7-2x}$ Specific Requirements

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6.5.2 O-RU$_6$ Specific Requirements

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6.5.3 O-RU$_8$ Specific Requirements

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6.6 FHGW/FHM – Common Requirements

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6.7 FHGW/FHM – Split Option Specific Requirements

6.7.1 FHM/Switch for Split 7-2x, Specific Requirements

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6.7.2 FHGW$_{7\text{-}2x\text{->}8}$ Specific Requirements

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6.7.3 FHM/Switch for Split 8, Specific Requirements

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6.8 Integrated gNB-DU Requirements

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History

Date	Revision	Description
2023.07.18	04.00	Final version for publication Update on O-RU requirements to reflect WG11 outcomes on security
2022.07.19	03.00	Final version for publication. Added new section for Massive MIMO high level architecture and high-level reference architecture for Integrated gNB-DU Hardware Architecture (O-RU Functional Split 2)
2021.07.16	02.00	Final version for publication
2020.10.21	01.00	Final version for publication

